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ORIGINAL ARTICLE

Utilization of the Fordham Risk Screening Tool for violence risk assessment in an emergency department

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Abstract

Background: Violence is a critical problem in the emergency department (ED) and patients experiencing mental health crises are at greater violence risk; however, tools appropriate for assessing violence risk in the ED are limited. Our goal was to evaluate the utility of the Fordham Risk Screening Tool (FRST) in reliability assessing violence risk in adult ED patients with acute mental health crises through evaluation of test characteristics compared to a reference standard.

Methods: We evaluated performance of the FRST when used with a convenience sample of ED patients undergoing acute psychiatric evaluation. Participants underwent assessment with the FRST and an established reference standard, the Historical Clinical Risk Management-20, Version 3 (HCR-20 V3). Diagnostic performance was assessed through evaluation of test characteristics and area under the receiver operating characteristic curve (AUROC). Psychometric assessments examined the measurement properties of the FRST.

Results: A total of 105 participants were enrolled. In comparison to the reference standard, the AUROC for the predictive ability of the FRST was 0.88 (standard error 0.39, 95% confidence interval [CI] 0.81–0.96). Sensitivity was 84% (95% CI 69%–94%) while specificity was 93% (95% CI 83%–98%). The positive predictive value was 87% (95% CI 73%–94%) and negative predictive value was 91% (95% CI 83%–86%). Psychometric analyses provided reliability and validity evidence for the FRST when used in the ED setting.

Conclusions: These findings support the potential utility of the FRST when used to assess violence risk in adult ED patients experiencing a mental health crisis. Future research with more diverse populations and ED settings is warranted.

KEYWORDS
crisis assessment, risk assessment, screening tool, violence

INTRODUCTION

Violent behavior toward health care workers is notably high.[1,2] Emergency departments (EDs), in particular, are known for their high rates of violence, with as many as 75% to 91% of ED nurses reporting experiencing violent incidents while at work.[3] These data are likely an underestimate given that many violent incidents in the health care setting go unreported.[4] For patients in mental health crisis, risk

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assessment for potential violence is essential, but is evaluated most often by unstructured assessment guided by clinical judgment.[5]

Structured tools to help clinicians evaluate violence risk do exist. The Historical Clinical Risk Management-20, Version 3 (HCR-20 V3), is a 20-item structured clinical judgment approach to assessing violence risk that has been used in over 200 empirical studies and is considered the criterion standard in the evaluation of potential risk for violence.[6,7] However, the utility of this well-validated tool has limited applicability in ED populations due to the several hours needed for completion.[8] In contrast, the Fordham Risk Screening Tool (FRST) was specifically designed to utilize information that should be readily available either before or during a standard psychiatric intake evaluation and can be completed in minutes.[9] Prior research on the validity of the FRST indicated that it is accurate and reliable for identifying psychiatrically hospitalized individuals who were subsequently evaluated to be at high risk for violence by the HCR-20 V3.[9,10] The FRST has, to date, no published data on its application in an ED setting. Therefore, we hypothesized that the FRST algorithm screening tool would demonstrate acceptable test characteristics (sensitivity, specificity, positive and negative predictive values, accuracy), to reliably identify patients at high risk for violence compared to the criterion standard violence risk screening tool, the HCR-20 V3 in patients presenting to the ED in mental health crisis.

METHODS

Design

We used a methodologic design to address our study objectives. Burns and Grove[11] describe methodological designs as being used to assess the validity and reliability of instruments that measure constructs such as violence risk; they are unique designs that specifically aid in instrument development or modification. Data were collected in a prospective, observational fashion. The study was exempted by the institutional review board at Maine Medical Center (MMC) and the requirement for written informed consent was waived. Instead, study participants were provided with a written participant information sheet providing details of the study and their roles and responsibilities as research participants, which did not require a signature.

Participants

Study participants ($N = 105$) were adult patients ages 18 to 65 years old who presented to the ED at MMC, were referred for crisis mental health evaluation, and were determined to require inpatient psychiatric admission after a standard psychiatric evaluation between June and August 2022. Patients ($n = 6$) who were not their own guardian, who refused or were unable to participate in comprehensive psychiatric evaluation, or who were unwilling or unable to provide informed consent were excluded from the sample. Participants were not provided with any financial remuneration for their participation.

Study procedures

Study personnel were composed of five psychiatrists and one medical student. Each received a standardized 3-hour training in utilization of the FRST screening tool given by one of the primary developers of the tool. The FRST algorithm, as described, is a screening tool based on known information about a patient's history of recent (defined as within the past 6 months) and severe (reflecting behavior that could have resulted in injury requiring medical evaluation) violent behavior, violent threats, and violent thoughts.[9,10] The length of time to perform the FRST administration in the ED environment was between 2 and 25 min. Study personnel also completed a standardized training (approximately 15 h in length) in the application and use of the HCR-20 V3. The HCR-20 V3[12] instructs the evaluator to critically consider which risk factors are most important to a particular patient's violence risk. The evaluator rates each of the 20 items as present, partial present, or absent. The evaluator then rates each item's relevance to the patient being evaluated as low, moderate, or high. While higher prevalence and relevance ratings typically indicate a greater likelihood of violence, clinicians utilize this tool to create a future violence risk/case prioritization formulation to make a final summary violence risk rating. The time for administration (including necessary chart review) of the HCR-20 V3 ranged from 45 to 180 min for administration in this study, with additional time to discuss case formulation and scenarios that are an important component of this tool. For the purposes of this validation study, results of both the FRST and the HCR-20 V3 did not affect clinical decision making.

Potential study participants were comprised of patients who presented to the ED and were referred for a mental health evaluation. A subset of those patients were determined by that evaluation to require inpatient psychiatric hospitalization. However, due to long wait times, lack of inpatient bed availability, and change in clinical status, only 63% of patients in the MMC ED who are referred for inpatient psychiatric hospitalization are actually hospitalized. This subset of patients, as described, were the pool of candidates for the study if they met all inclusion and exclusion criteria. Once identified, a member of the study team completed a FRST algorithm for each patient. Subsequently, after completion of the FRST algorithm, a different study investigator who was blinded to the results of the FRST, completed an HCR-20 V3 evaluation based on chart review and clinical interview. These evaluations were completed in a private space in the ED within 48 h of presentation to the ED. As these tools were completed in the ED, there were often times in which completion of both the FRST and the HCR-20 V3 were interrupted. In the case of interruption, study team members made attempts to complete the study interviews as close to the initial interruption as possible and were able to do so in all cases within a few hours of initiation of the interview. As the FRST had not been validated among this study population, results of the FRST were not shared with clinical staff and were not used in disposition decision making. However, to protect the safety of clinical staff, staff was alerted if patients rated on the HCR-20 V3 as high risk for imminent violence. To establish reliability in rating

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scales among study team members, the PI separately completed both a FRST and HCR-20 V3 evaluation on the first three evaluations of each tool completed by other team members.

Primary data analysis

Data were downloaded from a REDCap database and were analyzed using SPSS for Windows v. 28.0 (IBM Corp) statistical software. Descriptive statistics were used to summarize the characteristics of the study participants. We created a contingency table to summarize violence risk predictions using both the FRST and the HCR-20 V3. Standard analyses of the FRST's sensitivity, specificity, positive and negative predictive values, and accuracy were conducted to evaluate the instrument's potential utility as an assessment tool.[13] Given previously established reliability and validity in evaluating the risk of violence in adult patients, we used the HCR-20 V3 as the comparative standard in these analyses.[6] Estimation of the area under the receiver operating characteristic curve (AUROC), and its associated 95% confidence interval (CI), was used to assess the predictive performance of the FRST tool, after collapsing the HCR-20 V3 "Future Violence Case Prioritization" determination into a binary (low, low-moderate vs. moderate, moderate-high, and high) variable. Based on established recommendations for a sample of 10 subjects per item under evaluation, we determined that a minimum sample of 80 participants would be required to support our planned analyses.[14,15]

Sensitivity analysis

To assess for differences in functioning of the FRST, we repeated assessments of sensitivity, specificity, positive and negative predictive values, accuracy, and the AUROC based on participants' legal status: voluntary versus involuntary admissions. For these analyses, the HCR-20 V3 was again used as the reference standard.

RESULTS

Characteristics of study subjects

A total of 105 patients were enrolled in the study during a 3-month recruitment period; there were no missing data and all cases were included in the analysis. The mean age of the participants was 41 years (range 19–65 years). The majority of participants identified as male ($n=61$, 58%), followed by female ($n=38$, 36%) and transgender or nonbinary ($n=6$, 6%). Consistent with the demographics of our locale, most participants identified as White ($n=97$, 92%), followed by Black or African American ($n=6$, 6%) and Native American or Alaska Native ($n=2$, 2%). Six percent ($n=6$) of participants identified as Hispanic or Latine. Full demographic characteristics are provided in Table 1.

TABLE 1 Demographic characteristics of the study participants.

Characteristic	Full cohort ($N=105$)
Age (years)	40.5 (±12.7)
Gender	
Male	61 (58)
Female	38 (36)
Transgender or nonbinary	6 (6)
Race	
White	97 (92)
Black or African American	6 (6)
Native American or Alaska Native	2 (2)
Ethnicity	
Non-Hispanic or Latine	99 (94)
Hispanic or Latine	6 (6)
Marital status	
Single	68 (65)
Married or domestic partnership	13 (12)
Widowed	2 (2)
Separated	3 (3)
Divorced	19 (18)
Legal status	
Voluntary	86 (82)
Involuntary	19 (18)

Note: Data are reported as mean (±SD) or *n* (%).

Main results

The HCR-20 V3 identified 42 participants (40%) as being at low risk for future violence, 25 (24%) as being at low-moderate risk, 18 (17%) as being at moderate risk, 13 (12%) as being at moderate-high risk, and seven (7%) as being at high risk for future violence using the case prioritization method. We collapsed these categories for analysis (low, low-moderate vs. moderate, moderate-high, and high), resulting in 67 participants (64%) with a negative HCR-20 V3 and 38 (36%) with a positive HCR-20 V3. Application of the FRST tool identified 37 participants (35%) as being at higher potential risk for violence and requiring a more comprehensive violence screening (Table 2). Six participants with a negative FRST assessment were rated as positive on the HCR-20 V3: five participants who received a "moderate" future violence prediction designation and one participant who received a "moderate-high" designation with the HCR-20 V3. No participants with a negative FRST assessment screened positive for imminent violence ("moderate-high" or "high") on the HCR-20 V3 and one participant with a negative FRST screened positive for serious physical harm ("moderate-high" risk) on HCR-20 V3 assessment.

With the HCR-20 V3 as the reference standard, the AUROC for the predictive ability of the FRST was 0.88 (standard error [SE] 0.39, 95% CI 0.81–0.96, $p<0.001$). Figure 1 displays the associated ROC for the predictive ability of the FRST compared to the HCR-20 V3.

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TABLE 2 Contingency table for FRST and HCR-20 V3 measures ($N = 105$).

FRST comprehensive violence screen required	HCR-20 vs. 3 future violence case prioritization		
	Positive	Negative	Total
Positive	32	5	37
Negative	6	62	68
Total	38	67	105

Abbreviations: FRST, Fordham Risk Screening Tool; HCR-20 V3, Historical Clinical Risk Management-20, Version 3.

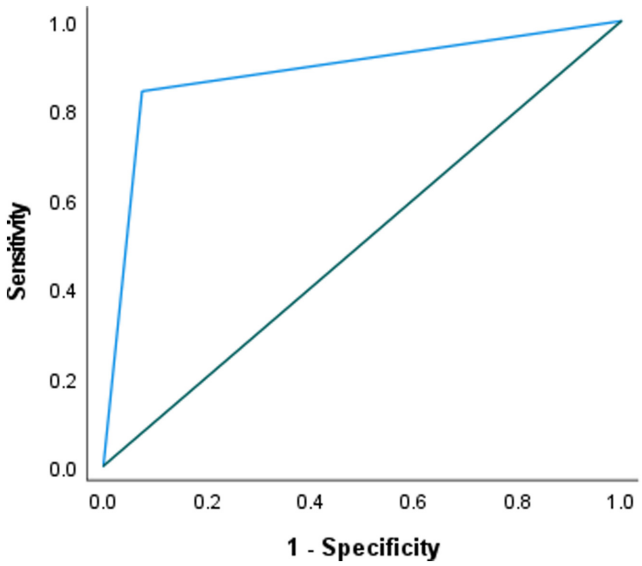

FIGURE 1 Recievier operating characteristic curve for the predictable ability of the FRST in comparison to the HCR-20, *v3* reference standard.

Sensitivity was 84.2% (95% CI 68.8%–94.0%) while specificity was 92.5% (95% CI 83.4%–97.5%). The accuracy of the FRST was 89.5% (95% CI 82.0%–94.7%). The positive predictive value was 86.5% (95% CI 73.1%–93.8%) and the negative predictive value was 91.3% (95% CI 83.2%–95.6%).

Reliability analyses

Internal consistency reliability of the FRST, as measured by the KR-20, was 0.79. Corrected item-total correlations (item discrimination) ranged from 0.46 to 0.62, all above the 0.30 threshold for deletion (Table 3). Interitem correlation values ranged from 0.227 (for the "violent thoughts severe" and "violent behavior recent" item pair) to 0.508 (for the "violent behavior recent" and "violent threats recent" item pair). Item difficulty values for the FRST data ranged from 0.23 (for the "violent threats recent" item) to 0.39 (for the "violent threats severe" item), representing endorsement of individual items in 23% to 39% of study participants.

TABLE 3 Item discrimination and difficulty values for the FRST instrument, six items ($N = 105$).

Item	Item difficulty	Corrected item discriminations	Alpha if item deleted
Violent behavior recent	0.24	0.46	0.77
Violent behavior severe	0.39	0.53	0.75
Violent threats recent	0.23	0.58	0.74
Violent threats severe	0.40	0.50	0.76
Violent thoughts recent	0.25	0.62	0.73
Violent thoughts severe	0.33	0.54	0.75

Abbreviation: FRST, Fordham Risk Screening Tool.

TABLE 4 Component loadings and communalities for the FRST tool ($N = 105$).

Item	Component 1 loadings	Communalities
Violent behavior recent	0.629	0.396
Violent behavior severe	0.687	0.472
Violent threats recent	0.733	0.538
Violent threats severe	0.657	0.428
Violent thoughts recent	0.771	0.594
Violent thoughts severe	0.699	0.488
Total extracted common variance 49%		
Total scale alpha (KR-20) 0.79		

Abbreviation: FRST, Fordham Risk Screening Tool.

Principal components analysis

Bartlett's test of sphericity ($\chi^2 = 153.784$, $df = 15$, $p < 0.001$) indicated that the correlation matrix for the FRST data was not an identity matrix and that correlations between the items were sufficiently large for factor analysis.[16] The Kaiser-Meyer-Olkin measure verified the sampling adequacy for the analysis and indicated that the magnitudes of the squared partial correlations tended to be small, 0.759.[17] The determinant of the correlation matrix was nonzero (0.192), indicating that multicollinearity was not a problem in these data. Given these results, we concluded that the correlation matrix for the FRST data contained sufficient variability for factoring.

Exploratory principle components analysis revealed a parsimonious single-component solution accounting for 49% of observed variance. All items loaded highly onto a single component, as shown in Table 4. Examination of the associated scree plot (Figure 2) also supported a single-component solution through demonstration of discontinuity in the plotted eigenvalues.[18] In addition, communality

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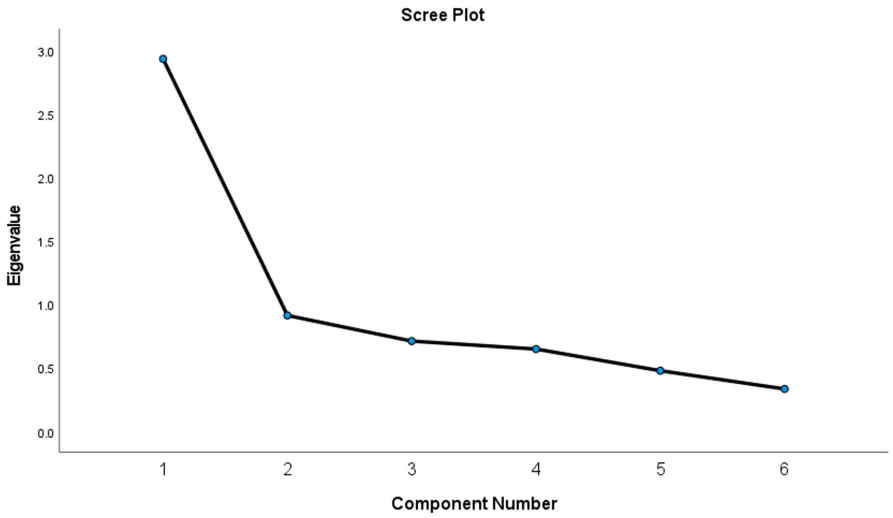

FIGURE 2 Scree plot of eigenvalues for the FRST items, $N = 105$.

values reflecting the common variance estimated for the sample are provided. These communality values represent the variance explained by the retained component following extraction. Values for the FRST items ranged from 0.396 to 0.594, accounting for a moderate proportion of the common variance. Given the single-component nature of the solution, rotation was not required to aid in interpretation of the solution.

TABLE 5 FRST comprehensive screening by legal status.

FRST	Total	Involuntary	Voluntary
Negative	68 (65)	9 (47)	59 (69)
Positive	37 (35)	10 (53)	27 (31)

Note: Data are reported as *n* (%).
Abbreviation: FRST, Fordham Risk Screening Tool.

Sensitivity analyses

Nineteen participants (18%) were being admitted on an involuntary basis while 86 (82%) were voluntary. As shown in Table 5, a greater proportion of involuntary admission participants screened positive on the FRST compared to their counterparts being admitted voluntarily (53% vs. 31%, respectively). In addition, differences were noted when comparing sensitivity, specificity, predictive values, and accuracy for these two groups (Table S1). When comparing the AUROCs for the predictive ability of the FRST compared to the HCR-20 V3 for these participant groups, a significant difference was noted as depicted in Figure 3 (AUC difference 0.314 [SE 0.142], 95% CI 0.036–0.593, $z = 2.210$, $p = 0.027$). These findings should be interpreted with caution due to the small number of participants admitted involuntarily.

DISCUSSION

Violence risk assessment has become increasingly recognized as an important component of the evaluation of patients who present to the ED in mental health crisis. While only a minority of patients with behavioral health concerns engage in violent behavior, those with acute exacerbations of their severe mental illness are at higher risk for violence.[19] State civil commitment laws link violence risk to involuntary hospitalization.[20] Because of this, emergency medical practitioners have been charged with violence risk assessment as a function of their evaluation and treatment of patients who present to the ED. A complete assessment of this risk requires an evidence-based, structured approach to maximize reliability. The added challenge of completing comprehensive assessments in the fast-paced environment of the ED lends itself to the use of evidence-based screening tools to optimize utilization of limited clinical resources. This study, therefore, sought to validate a screening tool, the FRST, in the ED setting as a means of identifying those patients with elevated potential for violent behavior. Overall, we found the results encouraging, suggesting that the FRST could be adopted for purposes of violence risk screening in an ED environment.

The traditional Classical Test Theory–based assessments of the psychometric properties of the FRST algorithm reported here provide evidence supporting the reliability of the instrument when used with adult patients experiencing mental health crises in the ED setting. Evidence supporting internal consistency reliability was provided by a standardized alpha coefficient of 0.8 for the FRST items with corrected item-total correlations that were above 0.30, indicating acceptable internal consistency.[11] Assessment of the AUROC for the predictive ability of the FRST in comparison of to the HCR-20 V3 (0.88 [SE 0.39], 95% CI 0.81–0.96, $p < 0.001$) provides evidence supporting concurrent validity. Further, factor analysis findings supporting a single-factor structure accounting for a good proportion of common variance help to affirm that the FRST, used in our setting

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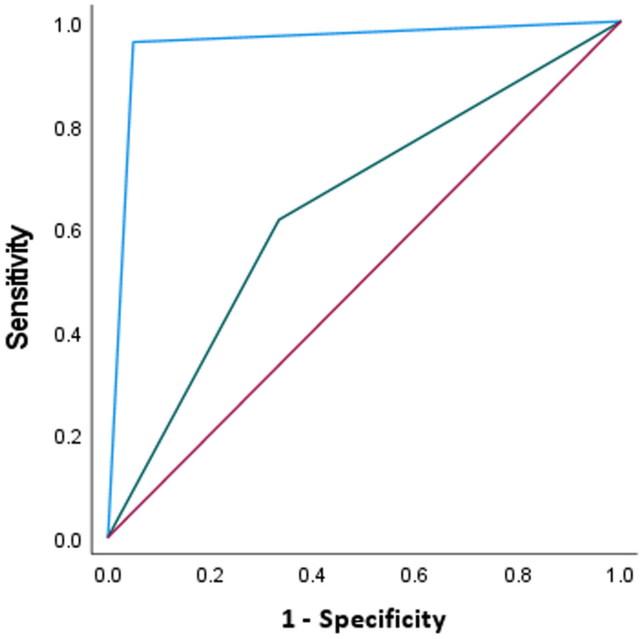

FIGURE 3 Comparison of the area under the recievier operating characteristic curves for the predictive abilityof the FRST by legal status. The blue line represents the reciever operating characteristic curve for the voluntary admission group while the green line depicts the ROC curve for the involuntary admission group. The area under the recieving operating characteristic curve for the predictive ability of the FRST as compared to the HCR-20, *v3* in the voluntary admission group was 0.936 (SE 0.037), 95% CI: 0.864–0.999, $p < 0.001$. The area under the reciever operating characteristic curve for the predictive ability of the FRST as compared to the HCR-20, *v3* in the involuntary group was 0.622 (SE 0.133), 95% CI: 0.362–0.882, $p = 0.369$.

and population, is measuring the construct it intended to measure, supporting construct validity.

Six participants had a negative FRST screen and positive HCR-20 V3 case prioritization, which amounted to 5.7% of the overall sample and 16.2% of the patients who were ultimately determined to be positive on the HCR-20 V3. Of those six participants, five had a moderate designation and one had a moderate-high designation on the HCR-20 V3. No patients were rated as high risk on the HCR-20 V3 who did not screen positive on the FRST. Given the importance of violent behavior, the authors chose to include all patients who were evaluated as moderate, on the HCR-20 V3 as a positive result. However, if patients who scored as moderate on the HCR-20 V3 are considered negative, then the sensitivity of the screening tool increases to 97%. The authors chose, instead, to take a conservative approach to these categorizations, thereby increasing the number of apparent false-negative participants.

A small subset of our participants ($n = 19$, 18%) were on an involuntary legal status, and sensitivity analyses were focused on assessing for differences in FRST functioning based on participant legal status. From a clinical perspective, one might logically expect a higher proportion of positive FRST outcomes among involuntary participants given that recent violent behavior or threats could constitute criteria for application for involuntary admission on the grounds of risk to others. Therefore, it was not surprising that a larger proportion of involuntary patients ($n = 10$, 53%) were identified by the FRST as being at potential higher risk for violence and requiring further evaluation when compared to their counterparts being admitted on a voluntary basis ($n = 27$, 31%). While findings should be interpreted with caution due to the small number of involuntary hospitalization participants, we were surprised at the lower point estimates for the sensitivity, specificity, positive and negative predictive values, and accuracy of the FRST in the involuntary group (Table 5). Further, the AUROC was much lower in the involuntary group (Figure 3), with a significant difference when comparing AUROCs for involuntary versus voluntary patients. These findings should be considered preliminary due to small sample size and large CIs for the involuntary group; however, in future studies with more robust samples, attention should be paid to differences in FRST outcomes stratified by legal status given the absolute differences observed here.

In this study, 35% of patients were screened by the FRST algorithm as requiring more comprehensive violence risk assessment. This is consistent with previous research published by Rosenfeld et al.[9] identifying 35.9% and Rotter and Rosenfeld[10] identifying 37.8% needing further violence risk assessment using the original FRST algorithm. This study, however, demonstrated superior sensitivity and comparable specificity to these previous publications. Given the limited information often available in an ED setting, it is somewhat surprising to find this degree of reliability on this screening tool. It is possible that the limited availability of information necessary to complete the HCR-20 V3 may have created an environment in which case-prioritizing methods were similar to the amount of information utilized to complete the FRST algorithm. Our assessments were also completed in serial, often in the same day, while prior studies had a more significant time gap between assessments,[9,10] which may have resulted in higher correlation between FRST and HCR-20 V3 results seen here.

LIMITATIONS

There are some notable limitations to this study. Given the rapid movement of patients in behavioral health crisis in the ED environment and limited number of research clinicians, it was not possible to capture consecutive patients who met inclusion criteria. There were a subset of patients who were transferred out of the ED environment prior to being approached to participate in the study. The ongoing nature of nursing and medical care in an ED environment also created conditions in which many patient interviews were temporarily interrupted. In addition, six patients who met inclusion criteria for this study either declined to participate or were unable to provide valid consent due to their clinical condition.

However, we believe that the fact that this research was completed within the confines of a traditional ED environment reflects its applicability to "real-world" ED settings. In addition,

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we are pleased that we were able to have meaningful consent discussions with the large majority of potential participants that we approached about the study, demonstrating that it is possible to enroll patients with acute mental health needs in clinical studies in the ED setting. This patient population is often overlooked in terms of research participation, limiting the availability of high-quality research evidence that is critical to providing a foundation for acute mental health care clinical practice in the ED. This study is additionally limited in generalizability due to the patient population being confined to one ED in Portland, Maine. Although there are no a priori reasons to suspect this population is markedly different than those throughout the United States in regard to violence risk specifically, our patient sample was largely male and White race (Table 1), so replication of similar studies geographically may elucidate any potential demographic-related differences.

CONCLUSIONS

For the purposes of this validation study, the results of the Fordham Risk Screening Tool were not used to impact clinical decision making. However, this study suggests that the Fordham Risk Screening Tool algorithm has favorable sensitivity, specificity, and psychometric properties when utilized as a screening tool for possible violence risk in the emergency setting. As medical facilities, and EDs in particular, seek out tools to address the epidemic of violence in both the community and the workplace environments, this study suggests that the FRST might serve as an important screening component of a more comprehensive violence risk assessment program. At the very least, this research adds to the body of literature supporting the FRST as a useful screening tool for violence risk assessment across additional treatment settings. Further research is necessary to establish the extent to which these results can be generalizable to other locations, hospital systems, and treatment settings. In particular, future research incorporating this tool into clinical practice by emergency medical practitioners is needed.

AUTHOR CONTRIBUTIONS

Christopher W. Racine, Linda S. Durst, Douglas N. Johnston, Tania D. Strout, and Lee A. Wolfrum conceived the study, designed the study, and obtained research funding. Christopher W. Racine, Tania D. Strout, Douglas N. Johnston, and Linda S. Durst supervised the conduct of the trial and data collection. Douglas N. Johnston, Ben J. Guido, Christopher W. Racine, Lee A. Wolfrum, and Kerry M. Quigley undertook recruitment of participating patients. Christopher W. Racine managed the data, including quality control. Tania D. Strout provided statistical advice on the study design, analyzed, and interpreted the data. Christopher W. Racine, Tania D. Strout, and Douglas N. Johnston drafted the manuscript, and all authors contributed substantially to its revision. All authors take responsibility for the paper as a whole.

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CONFLICT OF INTEREST STATEMENT

All authors report no conflicts of interest.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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